Lab 5 Mega Test — Set 1 Topics: NumPy operations, Pandas wrangling, Visualization, Linear/Logistic Regression **Time:** 60 min | **Total:** 100 marks

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Dataset

`customer_churn_data.csv` **Goal:** Predict `total_charges` (numeric) or `churn_status` (binary: 1/0) based on service usage features.

Part A — Data Wrangling & Feature Engineering (40 marks)

Q1. Outlier Detection & NumPy Logic [15 marks]

1. Load the dataset and focus on the `tenure_months` column.
2. Instead of simple clipping, use **NumPy** to calculate the mean and standard deviation. Identify any values that are more than 3 standard deviations away from the mean (outliers) and replace them with the median of the column.
3. Use `np.where` to create a new column `loyalty_flag` where 1 is assigned if `tenure_months` is above 48, and 0 otherwise.
4. Display the count of outliers found and the first 5 rows of the modified data.

Q2. Complex Mapping & Encoding [15 marks]

1. Create a column `contract_type_numeric` by mapping the `contract` column (e.g., 'Month-to-month', 'One year', 'Two year') to numerical values 1, 2, and 3 using a dictionary or mapping function.
2. For the `payment_method` column, perform **One-Hot Encoding** to convert categorical strings into separate binary columns.
3. Drop the original `payment_method` column and display the new DataFrame shape to confirm the expansion.

Q3. Conditional Cleaning & Normalization [10 marks]

1. Identify columns with more than 5% missing values and drop only those specific columns.
2. For the `service_type` column, use Pandas string methods to remove any leading/trailing whitespace and convert all text to uppercase to standardize the categories.
3. Display a summary of null values remaining in the dataset.

Part B — Statistical Analysis & Visualization (25 marks)

Q4. Multi-Level Pivot Analysis [12 marks]

1. Create a **pivot table** that calculates the **mean** and **variance** of `total_charges`.
2. Segment this data by both `contract_type` and `internet_service`.
3. Filter the pivot table to show only groups where the mean `total_charges` exceeds 500, and sort the results by variance in descending order.

Q5. Distribution Analysis via Boxplots [13 marks]

1. Generate a **boxplot** using Seaborn to compare the distribution of `total_charges` across different `churn_status` groups.
2. Customize the plot with a specific color palette (e.g., 'Set2') and add clear axis labels and a title.
3. **Short answer** (write in a markdown cell): Based on the boxplot, which group (Churn vs. No Churn) shows higher variability in spending? Why is understanding variance important before fitting a Linear Regression model?

Part C — Machine Learning Model (35 marks)

Q6. Advanced Prep & Scaling Logic [12 marks]

1. Define **X** (all relevant numeric features) and **y** (target: `churn_status`).
2. Apply **MinMaxScaler** to the features to squash values between 0 and 1.
3. **Short Question:** If the features have very different units (e.g., `tenure` in months vs. `total_charges` in thousands), why is scaling essential for the convergence of a Logistic Regression model?

Q7. Model Training & Evaluation [13 marks]

1. Split the data into 70% training and 30% testing.
2. Train a **Logistic Regression** model and predict the classes for the test set.
3. Display the **ROC-AUC Score** and a detailed **Classification Report**.

Q8. Comparative Evaluation [10 marks]

1. If the goal shifted to predicting the exact `total_charges`, explain why **Root Mean Squared Error (RMSE)** might be a "stricter" metric to use than **Mean Absolute Error (MAE)**.
2. Briefly describe how a high R-squared value combined with a high error on the test set might indicate **overfitting** in a regression model

Lab Mega Test — Deep Learning (PyTorch)

Set B **Time: 1 Hour**

Part A — PyTorch + DNN + Weighted Sampling (25 marks)

Q1. LFW People Dataset Classification using DNN

Use `sklearn.datasets.fetch_lfw_people(min_faces_per_person=70, resize=0.4)`.

1. Load the dataset `(X, y)`.
2. Normalize the features to `[0,1]`.
3. Create an imbalanced dataset by:
   - keeping only 25% samples from class 1
   - keeping all samples from remaining classes
4. Create: train = 70%, validation = 15%, test = 15% Using: stratification and fixed `random_state`
5. Convert the datasets into PyTorch tensors.
6. Create DataLoaders:
   - train → `batch_size=32`
   - validation/test → `batch_size=128`
7. Use: `WeightedRandomSampler` for the training DataLoader
8. Build a DNN: input → 512 → 128 → output
9. Use: ReLU activation, CrossEntropyLoss, Adam optimizer
10. Train for **5 epochs.**
11. After each epoch print: train accuracy and validation accuracy
12. Finally print: test accuracy

Part B — Data Augmentation + DataLoader + Visualization (20 marks)

Q2. STL10 Dataset Augmentation and Visualization

Use `torchvision.datasets.STL10`.

1. Create training transforms:
   - `RandomHorizontalFlip`
   - `RandomRotation(15)`

- `ColorJitter(brightness=0.2, contrast=0.2)`
- `ToTensor`

2. Create test transforms: `ToTensor` only
3. Create DataLoaders:
   - train → `batch_size=64`, `shuffle=True`
   - test → `batch_size=128`, `shuffle=False`
4. Pick one image (`idx=12`) and show: original image; augmented image using **2 subplots**
5. Print: shape of one image batch; shape of one label batch
6. Display: first 9 images from one training batch using a 3 × 3 grid.

Part C — Sequence Classification using Bi-LSTM (10 marks)

Q3. Disaster Tweet Classification using Bi-LSTM

Use the provided `tweets.csv` dataset containing: `text, target`

Where: `0` = non-disaster and 1 = disaster tweet

1. Load the dataset using `pandas`.
2. Perform: lowercase conversion, tokenization, vocabulary creation
3. Convert text into integer sequences.
4. Apply: padding/truncation (`max_len = 60`)
5. Build a Bi-LSTM model consisting of: Embedding layer, Bi-LSTM layer, Fully connected output layer
6. Train for **3 epochs.**
7. After each epoch print: training loss and training accuracy
8. Predict whether a custom tweet is: disaster and non-disaster

Part D — TensorBoard + Training Logs (10 marks)

Q4. TensorBoard Logging and Visualization

Using the model from Part A or Part C:

1. Train for **2 epochs.**
2. Log to TensorBoard: training loss and validation accuracy
3. Add: one batch of sample images/text
4. Store logs under: `./runs/`
5. Write the TensorBoard command required to launch TensorBoard.

Lab Mega Test 2

Part A — Supervised model + metrics (20 marks)

Q1. Decision Tree vs KNN on Breast Cancer Dataset

Use sklearn's Breast Cancer dataset (binary classification).

1. Load the dataset (X, y).
2. Split into train/test (80/20) with a fixed `random_state`.
3. Train the following models:
   - DecisionTreeClassifier (max_depth=5, fixed random_state)
   - KNeighborsClassifier (n_neighbors=5)
4. For both models, print:
   - Accuracy
   - Precision
   - Recall
   - F1-score
5. Plot confusion matrix for both models.
6. In 1–2 lines, compare their performance.

Part B — SVM + Cross-Validation (10 marks)

Q2. SVM with Cross-Validation

Using the same dataset:

1. Train an SVM model (SVC with kernel='rbf').
2. Perform **5-fold stratified cross-validation** using accuracy.
3. Print:
   - All 5 scores
   - Mean accuracy
   - Standard deviation

4. Compare (in one line) CV accuracy vs test accuracy.

Part C — Naive Bayes vs KNN (5 marks)

Q3. Model Comparison

Using the same train/test split from Q1:

1. Train:
   - GaussianNB
   - KNN (same as before)
2. Print test accuracy for both models
3. In one line, mention which model performed better and why.

Part D — Boosting (5 marks)

Q4. AdaBoost Performance

Using the same dataset:

1. Train AdaBoostClassifier with:
   - base estimator = decision stump (max_depth=1)
   - n_estimators = 100
2. Print test accuracy

University of Asia Pacific
Department of Computer Science and Engineering
Lab Final Examination, Fall 2025
Program: B.Sc. in CSE
4^{th} year 1^{st} Semester

Course Title: Operating System Lab
Course Code: CSE 402
Time: 1:5 hours
Credit Hour: 1.50
Full Marks: 40

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

Question 1: [20]

A computer system uses the Least Recently Used (LRU) page replacement algorithm with three page frames available in memory. Given the following page reference string:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3 —

Determine the number of page faults that will occur and show the state of the page frames after each reference. Assume that all frames are initially empty and that pages are loaded one at a time.

Question 2: [20]

In an Operating System, process management and synchronization are two important concepts used to control program execution and resource sharing.

Write a program to demonstrate the implementation of the exec() system call and semaphore synchronization mechanism.

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Your program should:

- Create a child process using fork().
- Use the exec() function in the child process to replace the current process with another program.
- Implement semaphore operations to synchronize access to a shared resource between processes or threads.
- Display appropriate messages showing process creation execution replacement and synchronization behavior.

7	0	1	2	0	3	0	4	2	3
7	7	7	2	2	2	2	4	4	4

Fault = 8